



Which factors matter in data-driven peer learning: Social intimacy, proficiency gaps, or selection autonomy?

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ABSTRACT

Peer learning is a promising strategy that encourages knowledge co-construction and reflection, while biased partner choice may hinder effective collaboration. Data-driven systems with interpretable learning analytics create new affordances for optimizing peer selection through knowledge awareness. Grounded in a technology-enhanced learning environment, this study investigates how social intimacy, knowledge proficiency gaps, and partner-selection autonomy affect peer learning in a Japanese university academic reading course. The data-driven conditions that underscored proficiency gaps in partner selection mitigated the influence of social intimacy and improved efficiency without compromising learning outcomes, although they alone were insufficient to inherently enhance engagement or learning quality. Helper-initiated pairing demonstrated relatively higher efficiency when learners could autonomously identify knowledge gaps and select partners. This study contributes a substantial implementation of computer-supported peer learning, an empirical analysis of key determinants of peer interaction, and a pedagogically aligned framework for integrating data-driven support into higher education to promote collaborative learning experiences. It also leads to future exploration of balancing these factors through contextualized design and continuously updated learner models incorporating multimodal data, thus extending applications to broader educational contexts.

1. Introduction

In higher education, flipped and collaborative learning have become increasingly accessible [1], as learners tend to be more self-regulated, cognitively capable, and socially independent [2]. These pedagogical approaches empower students to act as active agents rather than passive recipients of knowledge from monologue lectures, thus fostering soft skills that enable them to proactively identify and solve problems [3], and are increasingly promoted in the humanities and social sciences [4].

As an effective collaborative instructional strategy, peer learning can alleviate teachers' workload, particularly in large and flexible university courses where instructors may not be familiar with all students or able to monitor everyone during collaborative activities [5,6]. By encouraging meaningful interaction among learners, peer learning facilitates formative feedback and supports holistic personal development, aligning with the goals of sustainable education that underscore well-being and human creativity [7].

However, in practice, collaborative learning often involves biases, especially in partner determination. Most previous studies have concentrated on grading and feedback processes, paying less attention to how learning partners are selected [8]. Improper partner alignment can undermine the quality of peer interactions and feedback [9], which may stem from mismatched levels of understanding and social proximity. Furthermore, social expectations and interpersonal relationships among acquaintances can also impose emotional and psychological stress on learners [10].

To bridge the former gaps, artificial intelligence (AI) and learning analytics (LA) have orchestrated today's technology-enhanced learning environments, creating new opportunities to address these challenges through data-informed and personalized interventions. Nevertheless, most existing studies remain situated within teacher-centered and learner-centered contexts, while less extended to peer-centered learning settings [6]. Accordingly, empirical evidence remains insufficient on how theoretically relevant factors moderate the effects of data-driven partner-selection support on peer learning.

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The present study hence aims to clarify how social intimacy, proficiency gaps, and learners' autonomy shape peer learning processes and outcomes in a Japanese university academic reading course. It contributes by: (i) implementing a data-driven peer learning system that integrates the computational learner modeling with peer learning tasks in an authentic higher education context to demonstrate its practical applicability; (ii) proposing a comparative and theory-testing framework that examines social intimacy, proficiency gaps, and autonomy in partner selection through three distinct conditions; and (iii) showcasing how data-driven peer learning can be embedded into the pedagogical design and course schedule of a reading-focused university course to support structured peer review and revision activities.

2. Literature review

2.1. Peer learning and partner matching

Collaborative learning is a widely adopted instructional strategy that supports the development of interpersonal and socio-emotional competencies in contemporary society [11–13]. Peer learning, a specific form of collaborative learning in which learners engage directly with one another to review peers' work, identify gaps, and co-construct understanding [14,15], helps overcome certain limitations in large-group collaboration such as social loafing and free-riding [16]. It is rather promising in higher education contexts, where large class sizes, limited familiarity among students, and high learner mobility make it difficult for teachers to provide direct support [6,17].

By exposing learners to peers with diverse strengths, peer learning embodies the metaphor of constructivism that learning occurs when individuals actively construct meaning by linking prior knowledge with new information [18]. It also aligns with the perspective that knowledge is co-constructed through social interaction [19]. Further, the zone of proximal development (ZPD) theory [20] suggests the cognitive dimension of peer learning by emphasizing appropriate knowledge gaps between learners. The theory highlighted that learning is most effective when learners interact with peers who possess slightly higher competence levels. Hence, differences in knowledge proficiency and cognitive complementarity serve as crucial factors in determining optimal peer pairings.

Peer review is a seminal form of peer learning. It encourages learners to critically evaluate one another's work, provide constructive and formative feedback, and collaboratively refine their understanding [14,21]. Both providing and receiving feedback benefit learners [22]. Through this process, cognitive development, metacognitive reflection, and non-cognitive skills are fostered, promoting mutual benefit and knowledge co-construction. However, some challenges remain, such as psychological anxiety and the varying capability of participants to provide and explicate peer feedback [23,24]. Students may also reject peer feedback when they perceive peers as lacking sufficient knowledge or experience, making perceived expertise a determinant for feedback uptake [25].

Similar to the importance of group formation in orchestrating multi-member group work [26], determining appropriate peer partners affects the orchestration of effective peer learning. According to the ZPD framework, differences in knowledge and performance are often primary considerations in peer recommendation [27]. However, beyond cognitive aspects, as peer learning inherently involves interpersonal interaction, social emotional factors such as friendship and trust play equally important roles in sustaining productive collaboration and communication [10,13,28]. Further, learner autonomy is foregrounded as a critical factor in peer selection within computer-supported smart learning environments [29]. Studies on peer interaction and cooperative learning [30] indicate that when learners have agency in choosing partners, they develop stronger social reasoning, empathy, and internally driven motivation to participate [31].

Finally, although the evolving surge in AI agents provides versatile and scalable potential for high-quality and instant feedback, interaction with real humans remains indispensable for social and emotional development in digitalized societies [8,32]. It was pointed out that human peer learning should not be replaced by AI. Instead, AI should serve as a complementary tool to support and enhance peer interaction [33]. Beyond knowledge acquisition, developing interpersonal skills that contribute to well-being, such as deeper social reasoning and empathy [7], constitutes a vital mission of higher education as well.

2.2. Data-driven designs and applications for education

The need for data-driven design arises from the inherent individual differences among learners and the knowledge acquisition process that occurs through diversified communication and interaction based on constructivist theory. These individual differences manifest across multiple dimensions. In digitalized society, where technology-enhanced learning (TEL) has expanded through various tools and platforms [6] and breaks the geographical barriers. Data-driven systems can capture and quantify these multidimensional learner characteristics to build models that describe each learner's profile [34].

In current digital learning systems, it has been noted that both effort and expertise contribute to the effectiveness of peer feedback [35], which are typically measured through behavioral engagement and performance grades. These data constitute learner models, which lay the foundation for LA to generate actionable insights to enhance learning and teaching.¹ Other data collected from daily learning activities include trace logs that represent individual learning processes [36], as well as social-emotional and interpersonal aspects extracted from interaction logs and social network analysis indicators such as centrality measures [37]. Based on such learner models, learning support systems become adaptive through the integration of intelligent algorithms, enabling tailored material recommendations and optimized group formation [38,39].

However, the learner models used for prior performance in peer selection remain coarse, often distinguishing only between high and low gradient levels [22] or using cluster-based qualitative categories [40]. Quantitative modeling of knowledge gaps for peer assignment remains underexplored. The cold-start problem is also prevalent in collaborative learning contexts, as interactive activities do not always occur in regular classes. Cyclical data accumulation paves the way for capturing learning activities that are often difficult to record in everyday learning logs. The Group Learning Orchestration Based on Evidence (GLOBE) framework for iterative team-based learning [26] embodies this idea by enabling the learner model to be continuously updated with new data across four phases — formation, orchestration, evaluation, and reflection — within ongoing digital group learning platforms. This iterative process, in turn, enhances the precision and reliability of LA.

2.3. Open learner model and social awareness

The idea of Open Learner Models (OLMs) has been developed to make learners' attributes visible to both students and instructors, turning learning data into interpretable information that supports reflection and informed action [41]. In individual learning contexts, through external representations of learners' internal states, OLMs promote metacognitive and self-regulatory skills of learners [42]. Bodily et al. [43] have further combined OLMs with learning analytics to connect interpretability with large-scale behavioral data. Underpinned by this foundation, the Open Knowledge and Learner Model (OKLM) extends the conventional OLM framework by embedding a knowledge-centered

¹ <https://www.solaresearch.org/about/what-is-learning-analytics/>.

interpretive layer [44]. This design enables knowledge-aware LA, allowing the system to move beyond displaying raw traces, and instead, to produce more interpretable, knowledge-based indicators that guide learners' decisions and encourage effective learning strategies [45,46].

The notion of awareness originates from the field of Computer-Supported Cooperative Work (CSCW), defined as “an understanding of the activities of others, which provides a context for your own activity” [47]. Knowledge awareness, which refers to recognizing others' knowledge or expertise [48], enables learners to identify complementary strengths among peers and promotes mutual support and communication that lead to deeper knowledge construction [27].

With OLM visualizations embedded in LA dashboards, awareness has been operationalized through the reference frames (RFs) that provide interpretive contexts for comparison [49]. Social reference frames enable learners to compare their own performance with that of their peers [50]. For instance, dashboards that align a learner's engagement patterns with those of previous MOOC completers can offer external references for self-monitoring and goal setting [51]. However, current applications of RFs with LA have primarily focused on supporting self-regulated learning in individual learning contexts [52] or on assisting teachers' reflection and instructional decision-making [53]. The potential of OLMs to facilitate peer learning processes, especially in optimizing partner matching, still requires further coverage. [6,8].

3. OLM-enhanced data-driven peer learning platform

Grounded in the theoretical foundations above, we proposed an OLM-enhanced, data-driven peer learning platform that operationalizes these concepts into practice. Underpinned by insights from constructivist, socio-cognitive, and data-driven perspectives, the platform aims to address the inadequate support for peer learning activities in existing data-driven, technologically enhanced LA systems.

3.1. Underlying data infrastructure and peer recommendation

A data-driven peer learning platform based on Jiang et al. [54] was employed in this study to support student autonomy in partner selection. Within the system, learners can assume two roles: askers who post questions, and helpers who provide responses. To guide the selection process, we embedded a recommendation mechanism that incorporates knowledge metadata and proficiency data according to the aforementioned OKLM paradigm [44]. As illustrated in Fig. 1, the OKLM schema consists of two types of nodes. Knowledge nodes (K) represent learning units embedded in resources, whereas learner nodes (L) represent individual students. Edges among knowledge nodes constitute the knowledge model, while weighted links between learner nodes and knowledge nodes constitute the learner model. The weight assigned to each learner-knowledge link is the proficiency value $P_{l,k}$, where l denotes a learner and k denotes a knowledge node.

The system aggregates raw behavior logs from digital learning platforms such as annotations in e-book reading into proficiency values for specific knowledge units. The knowledge units are further interconnected within the network of the domain, thereby composing knowledge models. Further, the learner model is iteratively updated as students alternate between asking and helping roles, with the system integrating both individual and peer-interaction gains. This recursive update mechanism promotes knowledge diffusion and equitable participation among learners [55].

Beyond traditional forums, this peer learning platform integrates the OKLM-based learner model into the partner selection process. When initiating a new interaction, students can select the corresponding knowledge nodes, and the system presents anonymized recommendations of candidates based on the target knowledge. The recommendation draws on the proficiency values derived from the learner model, with peers who demonstrate higher proficiency in the related knowledge units appearing in the higher-ranked positions in the candidate interface. The detailed computational procedure, including proficiency updating and distance-discounted ranking, is provided in Appendix A.

Table 1

Comparison of asker-initiated and helper-initiated modes.

	Asker-initiated selection	Helper-initiated selection
Initiation	Asker initiates a post and selects from recommended helpers	Asker initiates a post and waits for helpers to choose
Decision maker	Asker decides which helper to select	Helper decides which asker (post) to respond to
Awareness support	Asker inspects radar graphs comparing own proficiency with candidate helpers	Helper inspects radar graphs comparing own proficiency with each asker
Ordering principle	Candidates ranked by their proficiency in related knowledge units (displayed to asker)	Questions ranked by the helper's own proficiency in related knowledge units (displayed to helper)

3.2. Asker-initiated and helper-initiated peer selection with OKLM visualization

Another feature of the platform is its flexibility in candidate selection guided by OKLM visualizations. As an awareness tool, the system visualizes knowledge proficiency gaps between the asker and potential helpers through radar graphs generated from OKLM data. Each axis corresponds to a selected or closely related knowledge node, and the plotted value represents the normalized proficiency score on that node. By overlaying the asker's and candidate's profiles, the radar graph helps learners identify whether a candidate is broadly stronger or has a specific advantage on the knowledge unit needed for the current task, thereby tempering aversion to algorithmic opacity. By anonymizing the identities, the interface promotes objective, knowledge-based decision-making to minimize the influence of personal or social factors. This consideration aligns with the rule that technology should scaffold but not replace the internal mediation and decision process of human [8].

To examine the effect of selection autonomy in partner determination, we extended the selection module to operate in two modes. In the *asker-initiated* mode (Fig. 2), which corresponds to the original design of Jiang et al. [54], the asker checks the radar graphs of potential helpers and selects one when initiating a question. In the *helper-initiated* mode (Fig. 3), the asker simply posts a question, and helpers hold the autonomy to decide which question to answer. The helper inspects radar graphs that compare his or her proficiency with the anonymized asker for each question, and the system orders the candidates according to the helper's own proficiency in the related knowledge units. Table 1 gives a comparison of the two modes.

4. Research design and implementation

To examine how *social intimacy*, *proficiency gap*, and *partner-selection autonomy* shape peer learning dynamics in the data-driven environment introduced above, we designed this empirical study around three research questions following the analytical framework proposed by Janssen and Kirschner [56]. They address the antecedent, process, and consequence dimensions of peer interaction respectively.

RQ1 (Antecedent): How do social intimacy and proficiency gaps between pairs differ across the three pairing conditions?

RQ2 (Process): How do learners' behavioral engagement patterns differ across conditions?

RQ3 (Consequence): Do learning outcomes differ across conditions, as reflected in report grade gains and changes in report editing records?

To investigate these questions, peer learning pairs were organized under three conditions that varied in the autonomy and mechanism of partner selection:

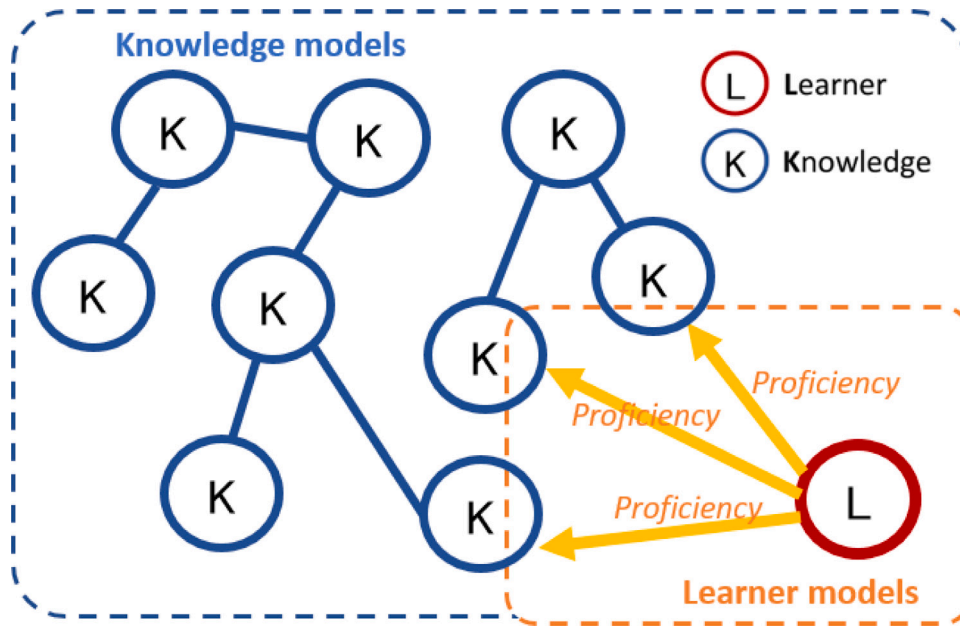


Fig. 1. The Open Knowledge and Learner Model (OKLM) paradigm. Knowledge nodes (K) represent domain units embedded in learning resources and are connected as a knowledge network, while learner nodes (L) are linked to these knowledge nodes via proficiency values derived from learning logs and evidence of peer interactions. The recommendation is generated by combining proficiency with the graph distance between the candidate's knowledge profile and the target knowledge nodes of the peer-learning request.

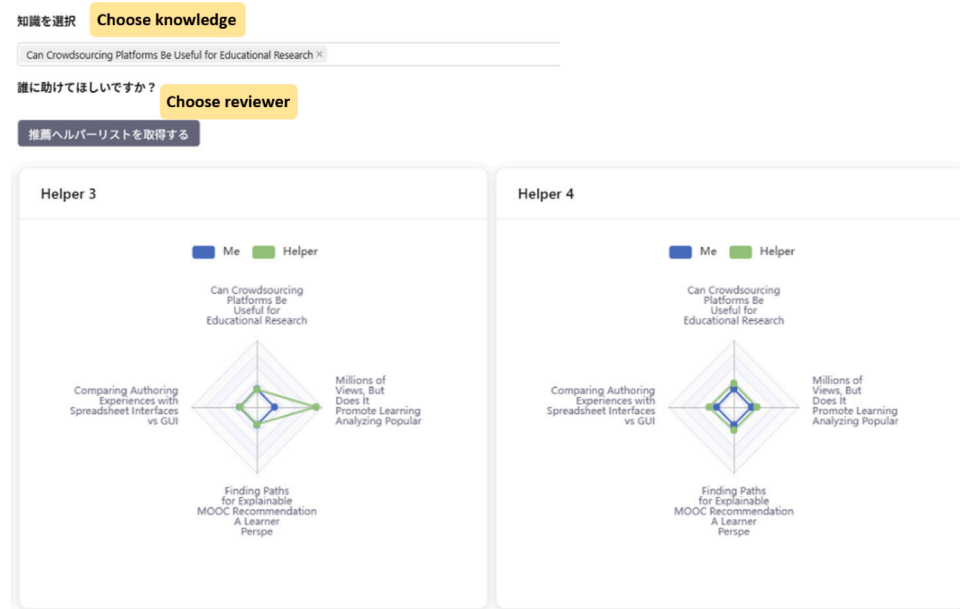


Fig. 2. The interface of asker-initiated selection mode.

- **Condition 1 (C1, Free Forum):** Helpers were chosen spontaneously through the Moodle forum without system recommendations.
- **Condition 2 (C2, Asker-initiated Selection):** The asker selected a reviewer (helper) from candidates recommended based on the OKLM paradigm (Fig. 2). The anonymized radar graphs allowed askers to compare their own proficiency with potential helpers within the same topic domain.
- **Condition 3 (C3, Helper-initiated Selection):** The reviewer (helper) selected a report (asker) to review (Fig. 3), also based on OKLM recommendations, reversing the direction of selection autonomy.

Accordingly, the study compares: (i) C1 (forum-based pairing) versus C2 and C3 (data-driven pairing) to analyze the roles of *social intimacy* and *proficiency gap*; and (ii) C2 versus C3 to examine the effect of *selection autonomy* in peer learning orchestration.

4.1. Participants

The study was conducted in an academic reading course at a national Japanese university during the 2025 academic year, targeting students above the sophomore level. Written informed consent, including data anonymization and ethical approval declarations, was obtained prior to data collection across systems. The course enrolled

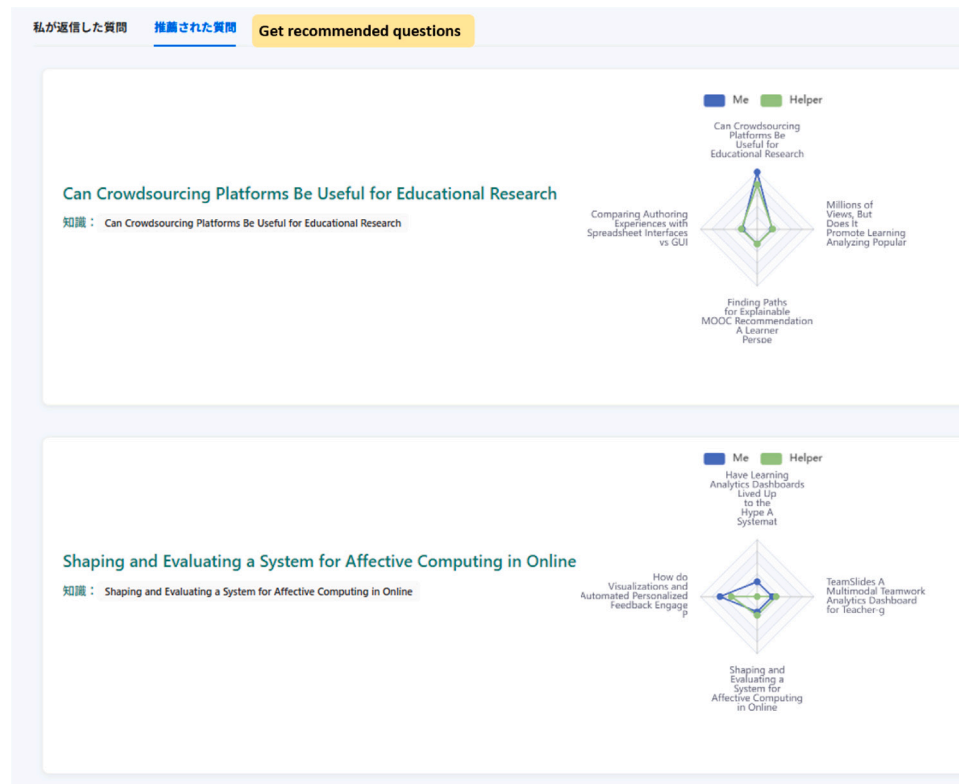


Fig. 3. The interface of helper-initiated selection mode.

34 students, of whom 28 completed all requirements and earned course credit. Most participants majored in different fields within the Department of Engineering, and two were from the Department of Agriculture.

4.2. Procedure and instructions

As a vital practice in the academic reading course, students engaged in paper reading and peer review activities during the latter half of the semester. Prior to this phase, they had developed basic academic reading skills by taking lectures and skimming assigned papers on the online platform and generating initial reading logs. The peer review activity consisted of three rounds over nine weeks and was designed as a within-subjects study where students were scheduled to experience all three conditions once. Due to absences or incomplete submissions, only completed cases in each round were included in the analysis. Each round of the peer review activity lasted for two weeks and consisted of three phases: report, review, and revision. In every round, students were required to select a new paper on which they had not previously written a report.

- **Report (Week 1):** Each student read one academic paper using the online e-book platform BookRoll [57] and wrote a two-page report summarizing and discussing its content.
- **Review (Week 2):** Each student then served as a reviewer (helper), reading a peer's report and providing written feedback. The asker-helper pairs were assigned according to the three experimental conditions described in the beginning of this section, and once assigned, pairs remained fixed for the duration of the round. Each student served as a helper only once in each round.
- **Revision (Week 2):** Report authors (askers) revised their drafts based on the feedback received and submitted a final version. They also evaluated the perceived helpfulness of the feedback on a five-point Likert scale.

4.3. Data collection

Social intimacy. Since social relationships are implicit and difficult to quantify, we measured social intimacy through a pre-test survey following Yuan et al. [58] and Zohar and Tenne-Gazit [59] that elicited self-reported friendship ties among students. In total, 20 friendship pairs were reported in the pre-survey, including 13 mutually acknowledged relations.

Please select all the people with whom you have talked about personal issues (e.g., good news about your personal life, something that troubles or frustrates you).

Proficiency gaps. The peer learning system was connected to a graph-based literature network following the OKLM schema. The papers and their interconnections were constructed from open-access articles published in the 14th Learning Analytics and Knowledge Conference (LAK24). Accordingly, a hierarchical two-layer knowledge network was built based on the session themes of each paper, as outlined in the official proceedings. A few papers that were not listed on the conference website were manually assigned to themes by the course instructor. The first-layer nodes represent research topics, and the second-layer nodes correspond to individual papers under each topic. Fig. 4 presents an overview of these topics (first-layer nodes), each of which includes several papers (second-layer nodes) used in the academic reading course.

We calculated raw proficiency scores (ranging from 0 to 1) following the data-driven design of the OKLM infrastructure introduced in [54]. Initial proficiency was collected from BookRoll, based on students' reading activities in the first half of the semester. Proficiency was subsequently updated with data from the peer learning system, reflecting the papers students wrote reports on and the papers they read or reviewed. Specifically, the updated scores integrated information from created annotations in BookRoll, the papers students had reported

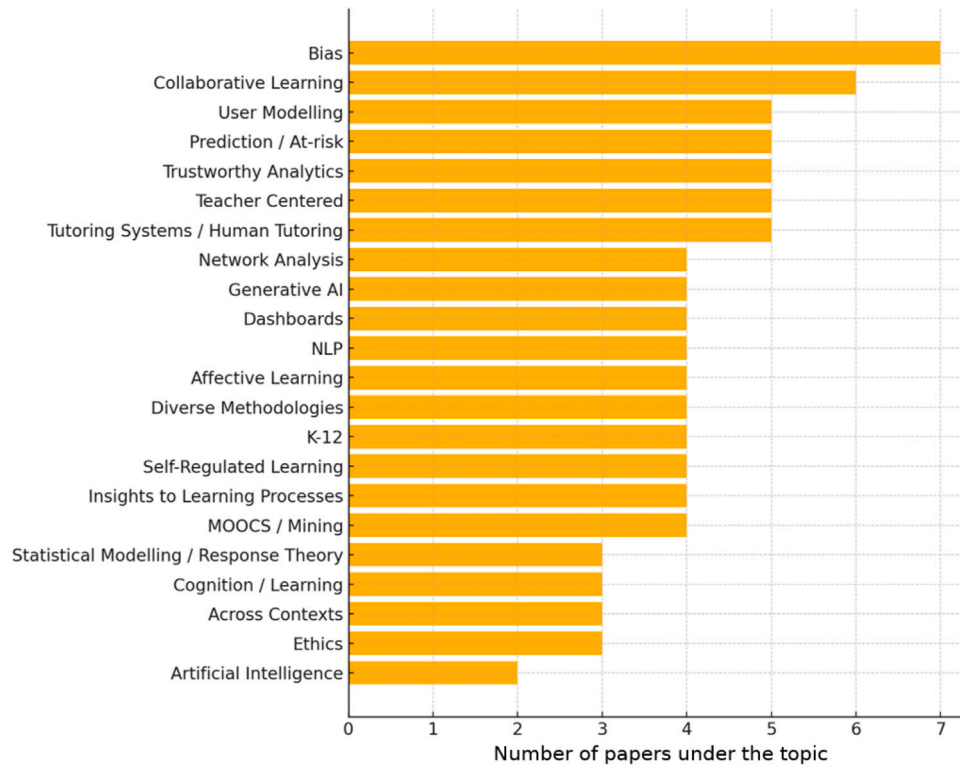


Fig. 4. Overview of the topics (first layer knowledge) and papers (second-layer knowledge) for the academic reading course.

on, and the papers they had reviewed. The detailed calculation of the study-specific proficiency indicators is provided in [Appendix A.3](#).

With regard to the proficiency gap, following the two-layer knowledge model, we computed two indicators representing the relative proficiency difference between the helper (H) and the asker (A). At the paper level, p_H and p_A denote the proficiency scores of the helper and asker, respectively. At the topic level, $\mathbf{H}$ and $\mathbf{A}$ represent their corresponding proficiency vectors based on paper–topic assignments. The two indicators are defined as follows:

$$\text{Diff}_{\text{paper}} = \frac{p_H - p_A}{p_H + p_A}, \in [-1, 1] \quad \text{Diff}_{\text{topic}} = \frac{\|\mathbf{H}\|_2 - \|\mathbf{A}\|_2}{\|\mathbf{H}\|_2 + \|\mathbf{A}\|_2}, \in [-1, 1] \quad (1)$$

Here, $\text{Diff}_{\text{paper}}$ represents the normalized difference in paper-level proficiency between the helper and the asker, while $\text{Diff}_{\text{topic}}$ captures their relative difference in topic-level proficiency, calculated using the L2 norm of their topic vectors. Positive values indicate that the helper has higher proficiency than the asker, whereas negative values indicate the opposite.

Behavioral engagement. The digital platforms used in this study recorded learner behaviors through embedded data sensors that generated xAPI statements in the *subject–verb–object* format. Each statement specified who performed which action on what content. Logs were extracted from BookRoll, the Moodle forum, and the peer learning platform after helpers had been assigned in the second week of each activity cycle and continued until the submission deadline of the final report. Thus, the trace data covered only the peer review and revision period, while behaviors related to individual reading and initial report writing were not included. All behaviors were then coded according to the schema shown in [Table 2](#). For clarity, we refer to behaviors from both the Moodle forum and the peer learning system as “peer interaction”.

Learning outcomes. We employed the reading reports to estimate learning outcomes, and only askers who attended the peer selection session and submitted both drafts were included in the analysis. Here, the initial report score refers to the score of the first report submitted before

peer feedback, whereas the revised report score refers to the score of the final version submitted after peer review and revision. As part of the course grade, the final reports were evaluated and graded by the teacher using the evaluation rubrics provided in the [Appendix](#), with sub-scores assigned for each rubric construct. To reduce potential bias in human grading across the two consecutive reports and obtain a more objective measure, we estimated learning gain by examining changes from initial report scores to revised report scores, which were graded by GPT-4o based on the same rubric and calibrated with several shots of the teacher’s grading in the prompt (see [Appendix B](#)). The consistency between automated and human ratings of final reports was examined using Wilcoxon signed-rank tests across the three rounds. No significant differences were found between the two sources (all $p > .05$), indicating that the automated scores aligned well with human judgments. Hence, using automated scoring results for subsequent analyses is considered valid and reliable.

We also analyzed word count differences between the initial and revised reports as an auxiliary indicator of students’ revision effort after receiving peer feedback, since text length can provide an accessible trace of engagement [35]. Only English words were counted, as the course syllabus required all reports to be written in English.

4.4. Data analysis

For RQ1, we conducted interpretive analyses of the distributions of social intimacy and proficiency gaps across the three conditions to confirm the presumed differences among them.

For RQ2, we first compared the overall behavioral engagement derived from xAPI logs collected from the two learning platforms across the three conditions using chi-square tests to obtain a general picture of engagement patterns. We then examined how social intimacy affected learner engagement in the free forum condition (C1), where an adequate number of friendship-driven pairings were observed. Since the engagement data were not normally distributed, Mann–Whitney U tests were conducted.

To inspect the effect of proficiency gap, we analyzed the two data-driven conditions (C2 and C3) and structured the dataset where each record represented one observation characterized by four potential moderating covariates: (a) learner *Role* (asker or helper), (b) *System* used (BookRoll or peer learning), (c) knowledge-node *Level* (paper or topic), and (d) selection *Initiator* (asker- or helper-initiated). The dependent variable was engagement, and the predictor was the proficiency gap ranging from -1 to 1 , as defined in Section 4.3. As multiple observations were nested within each helper–asker pair, we conducted multiple linear regression with clustered standard errors at the pair level to account for within-pair dependence [60]. This regression approach was used to examine whether the association between proficiency gap and engagement varied according to theoretically relevant contextual factors embedded in the experimental design. Four parallel moderation models were estimated with identical structures, each replacing the moderator Z with *Role*, *System*, *Level*, or *Initiator*. In this model, β_1 indicates the baseline relationship between proficiency gap and engagement, β_2 captures the main effect of the moderator, and β_3 indicates the interaction effect. ϵ denotes the residual error.

$$\text{Engagement} = \beta_0 + \beta_1 \text{Diff} + \beta_2 Z + \beta_3 (\text{Diff} \times Z) + \epsilon. \quad (2)$$

Further, to explore temporal transitions between individual reading and peer interactions, we employed hourly heat maps to compare the proportions of individual reading (BookRoll) versus peer interactions within each hour slot, using chi-square tests assessing their distributional differences.

For RQ3, since the homogeneity of regression test between the initial and revised report scores required for ANCOVA was violated ($F = 11.69, p < .01$), a scatter plot (see Fig. 11) was generated to interpret the nature of this interaction. To mitigate the ceiling effect observed in the regression plot, normalized learning gains [61] were computed using the following equation to compare differences among the three conditions. Non-parametric Kruskal–Wallis H tests were used due to the non-normal distribution.

$$LG = \frac{\text{Revised report score} - \text{Initial report score}}{\text{Full score} - \text{Initial report score}} \quad (3)$$

Regarding changes in word count, because their distributions were highly skewed and deviated significantly from normality, non-parametric Kruskal–Wallis H tests were used to compare the three conditions. As the posted contents consisted of structured reports and reviews based on predefined prompts rather than open-ended discussions, further semantic analysis was not applicable.

5. Results

This section reports the results corresponding to the three research questions: (i) Peer selection patterns showed that spontaneous pairing in the forum tended to preserve existing friendships, whereas data-driven pairing generated pairs with fewer social ties but more proficiency gaps. (ii) Overall engagement declined under the data-driven conditions, and students invested less effort on rereading the paper. This disposition was more evident when helpers determined the pairing. Friends exhibited lower participation in peer interactions, and larger knowledge gaps particularly reduced askers' activity levels. (iii) Despite these behavioral differences, no significant variation was observed in learning gains or report length across conditions.

5.1. RQ1: Peer compositions

Social intimacy. Regarding social intimacy between pairs, students demonstrated varying levels of friendship ties across conditions (see Fig. 5). In Condition 1 (free forum-based choice), seven pairs (28%) included friendship ties, whereas the other eighteen pairs did not. In Condition 2 (asker-initiated selection), only two pairs (7.4%) involved

friendship ties, including one mutual friendship. In Condition 3 (helper-initiated selection), no pairs showed friendship ties. These results confirm that spontaneous peer selection in the forum tended to preserve existing friendships, while the data-driven platform generated pairs with fewer friendship connections.

Proficiency gap. We compared paper-level proficiency gaps across the three conditions (see Fig. 6). The Kruskal–Wallis test revealed no significant differences ($\chi^2 = 1.22, p = .543$), with roughly half of the pairs showing higher helper proficiency in each condition.

For topic-level proficiency gaps, the overall difference was marginally significant ($\chi^2 = 4.74, p = .093, \epsilon^2 = .035$), indicating a rather small effect. The proportion of pairs with higher helper proficiency increased notably from 40.0% in C1 to around 70% in C2 and C3. Pairwise Mann–Whitney U tests showed marginal trends, with C2 tending to be higher than C1 ($U = 232.0, p = .054, r_{tb} = .313$), and C3 tending to be higher than C1 ($U = 271.0, p = .080, r_{tb} = .277$). The contrast between C2 and C3 was not significant ($U = 447.0, p = .507$).

5.2. RQ2: Behavioral engagement and process analysis

5.2.1. Overall behavior engagement

We first compared behavioral engagement across the three conditions. The overall chi-square test revealed significant differences in behavioral distribution ($\chi^2(14) = 364.82, p < .001, V = .213$), indicating a small-to-medium effect. Post-hoc pairwise comparisons confirmed detailed variations. Between C1 and C2 ($\chi^2(7) = 136.27, p < .001, V = .213$) and between C1 and C3 ($\chi^2(7) = 189.54, p < .001, V = .251$), pronounced differences were observed in *highlight_br*, *read_br*, and *open_pl*. The comparison between C2 and C3 ($\chi^2(7) = 160.31, p < .001, V = .282$) showed that *read_br* and *other_br* occurred more frequently in C2, whereas *view_pl* was substantially higher in C3. The Cramér's V values indicated small-to-medium effects for the overall comparison and C1 versus C2 (both $V = .213$), and effects approaching medium magnitude for C1 versus C3 ($V = .251$) and C2 versus C3 ($V = .282$).

As shown in Table 2, BookRoll behaviors such as reading and highlighting declined sharply from C1 to C3 (*read_br*: 579 → 267 → 157; *highlight_br*: 115 → 28 → 0), suggesting reduced reliance on the e-book platform in later conditions. Meanwhile, peer-learning behaviors (*open_pl*, *reply_pl*, and *view_pl*) became relatively dominant, indicating a shift of engagement from individual reading to peer interaction. The increase in *view_pl* in C3 reflects greater participation in viewing and responding to peers' posts within the peer-learning platform.

5.2.2. Effect of social intimacy in the free forum

We examined whether friendship influenced behavioral engagement in C1. On BookRoll, students without friendship connections showed higher average reading log counts ($M = 28.4$) than those with friendships ($M = 20.2$). On the forum, students without friendships also showed higher participation ($M = 50.9$) compared to those with friendships ($M = 26.2$). Mann–Whitney U tests revealed no significant difference in BookRoll engagement ($U = 62.0, p = .311$), but a significant difference in peer learning engagement ($U = 81.5, p = .015 < .05, r_{tb} = .70$), with a large effect size. This result suggests that students with closer friendships tended to interact less actively in the online forum.

5.2.3. Effect of proficiency gaps in the data-driven peer learning platform

Moderation by Role. A significant *Role* × *Gap* interaction was observed ($\beta_3 = 6.61, p = .030$). Simple-slope analyses indicated that the gap-engagement relationship was significantly negative for askers ($\beta = -6.44, p = .020$) but non-significant for helpers ($\beta = 0.16, p = .917$). As shown in Fig. 7, askers' engagement decreased as the proficiency gap widened. Large ability differences discouraged lower-proficiency

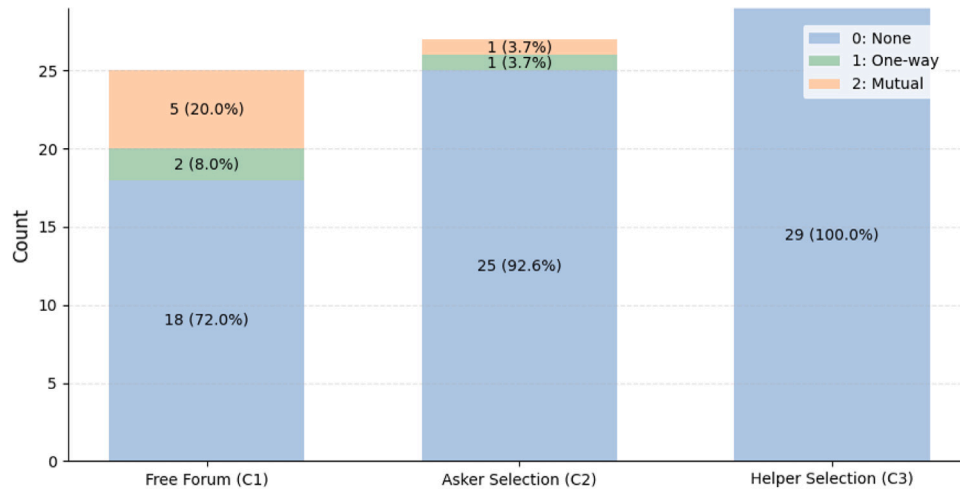


Fig. 5. Friendship ties for three conditions.

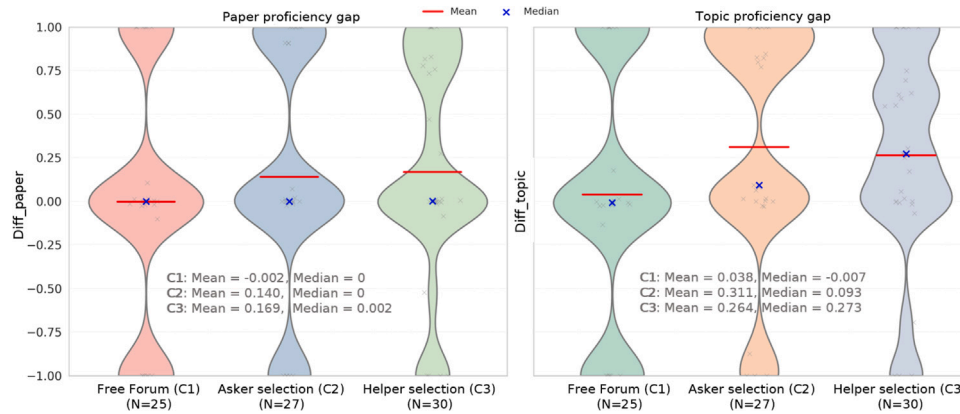


Fig. 6. Violin plots of knowledge gap distributions for three conditions.

Table 2
Behavior coding, descriptions, and frequencies across conditions.

Behavior Coding	Description	C1	C2	C3
BookRoll reading				
open_br	Opening the BookRoll system	98	79	72
read_br	Reading actions such as page turning or navigation	579	267	157
difficult_br	Highlighting unclear parts with a yellow marker	7	2	0
highlight_br	Highlighting important parts with a red marker	115	28	0
other_br	Other BookRoll operations (e.g., adding bookmarks)	77	82	28
Peer learning				
open_pl	Opening the peer learning system (forum or peer platform)	302	102	115
reply_pl	Replying to a peer's report post	82	53	49
view_pl	Viewing a peer's report post	747	398	590

learners from active participation, while helpers' engagement remained stable.

Moderation by System. The *System* $\times$ *Gap* interaction was not significant ($\beta_3 = 0.22$, $p = .950$). Simple slopes suggested a modest negative trend in BookRoll ($\beta = -3.25$, $p = .070$) and a weaker, non-significant trend in the peer learning system ($\beta = -3.03$, $p = .298$). The effect of knowledge gap on engagement was thus consistent across systems.

Moderation by Level. The *Level* $\times$ *Gap* interaction was not significant ($\beta_3 = 1.84$, $p = .176$). Still, the negative relationship between gap and engagement was stronger for paper-level proficiency ($\beta = -4.06$, $p = .018$) than for topic-level proficiency ($\beta = -2.22$, $p = .244$). A possible interpretation is that helpers with high paper-level proficiency are already familiar with the specific paper and therefore do not need to

re-check the original article during the peer review process. In contrast, when proficiency is derived from topic-level aggregates, the helper may still need to spend time revisiting the paper because it is relatively new to them.

Moderation by Initiator. The *Initiator* $\times$ *Gap* interaction was not significant ($\beta_3 = 1.21$, $p = .731$). However, the slope was significant in C2 (asker-initiated, $\beta = -3.69$, $p = .037$) but not in C3 (helper-initiated, $\beta = -2.49$, $p = .417$). This indicates that decision direction moderates the influence of proficiency differences: when weaker learners initiated pairing, large gaps reduced engagement, whereas when stronger learners initiated, the negative effect got weaker.

Across the four parallel models, only the *Role* $\times$ *Gap* interaction reached statistical significance, primarily driven by a sharp decline in

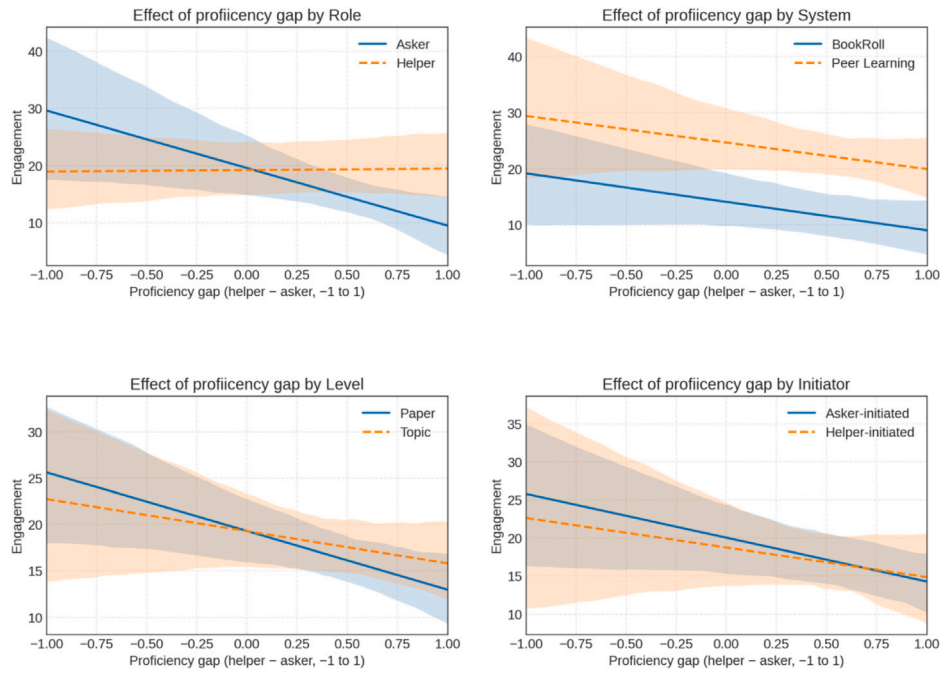


Fig. 7. Predicted engagement as a function of proficiency gap across four moderators. (a) Role, (b) System, (c) Level, (d) Initiator. Shaded areas represent 95% cluster-bootstrap confidence intervals.

engagement among askers as ability differences increased. The patterns for *Initiator* and *Level* showed a similar but weaker tendency, and differences in *System* were minimal. These results highlight asymmetric participation dynamics with proficiency gaps in peer learning that large ability gaps inhibit weaker learners' engagement and do not proportionally enhance helpers' engagement either.

5.2.4. Connection between BookRoll reading and peer interactions

We first conducted transition analyses using the lag sequence method and found that all significant behavioral transitions occurred within the same system. This result indicates a limited connection between the individual reading activities on BookRoll and the “review and revise” processes in peer learning platforms. Figs. 8 and 9 illustrate the hourly engagement of the two activities. We can see the engagement in both systems is the most intensive during the lecture when the task was assigned, and there are scattered engagement during the week before the next class.

To further examine this disconnection, we visualized learners' engagement distributions using hourly heat maps (Fig. 10). Each grid cell represents one hour, and its color intensity indicates the proportion of students who engaged only in peer learning activities among those active during that hour. This measure reflects how strongly peer learning activities dominated each hourly slot relative to reading activity. The slot-level proportion for each grid cell was defined as:

$$P_{\text{hour}} = \frac{N_{\text{students active only in peer-learning during the hour}}}{N_{\text{students active in any activity during the hour}}} \quad (4)$$

To summarize the overall tendency across learners, we further computed a global index based on the total number of grid cells across all participants, treating each learner-hour as one observation unit:

$$P_{\text{overall}} = \frac{N_{\text{peer-only hours}}}{N_{\text{peer-only hours}} + N_{\text{BookRoll-only hours}} + N_{\text{both-activity hours}}} \quad (5)$$

A chi-square test on these overall proportions revealed significant differences among the three conditions ($\chi^2(2) = 8.38, p = .015, V = .162$), indicating a small-to-medium effect. Post-hoc comparisons further indicated that the proportion of peer-learning-only cases was higher in C3 than in C1 ($\chi^2 = 7.02, p = .008 < .01, r = .178$),

while the difference between C1 and C2 was marginal ($\chi^2 = 2.86, p = .091, r = .114$), and the difference between C2 and C3 was not significant ($\chi^2 = 0.62, p = .431, r = .056$). These results suggest that students in C3 were significantly more likely to engage exclusively in the peer learning platform without concurrently using BookRoll, compared with those in C1. This pattern suggests that peer-based “review and revise” processes became more independent from individual reading activities in C2 and C3 where proficiency gaps determine the peer.

5.3. RQ3: Learning outcomes as consequence

Learning gains after review and revision. As shown in Fig. 11, students with higher initial report scores tended to cluster near the upper boundary of the scale in the revised report, especially for C2 and C3. This indicated a ceiling effect that limited the observable improvement among high-achieving learners. Table 3 shows the descriptive statistics of the normalized learning gain (LG) to alleviate the ceiling effect. The Kruskal–Wallis test revealed no significant difference among the three conditions ($H(2) = 4.34, p = .114, \epsilon^2 = .035$). The difference was not statistically significant, though the descriptive trend indicates that learners in the free forum condition achieved relatively higher improvement.

Although the gains in the report scores did not significantly differ across conditions, the process analysis in RQ2 suggested that the recommendation system reduced the effort students invested in selecting partners. Further, social intimacy (friendship ties) did not significantly affect learning outcomes. Correlations between proficiency gaps and learning gain remained weak, with coefficients around 0.1 or lower.

Text length variation. We also compared the change in word count between the first and second submissions across the three conditions. Descriptive statistics are shown in Table 4. The Kruskal–Wallis test showed a marginal but not significant trend ($H(2) = 3.90, p = .142 > .05, \epsilon^2 = .028$). Nevertheless, the descriptive pattern suggests that students in C3 tended to exhibit relatively larger gains in word count compared with the other two conditions. The median change was slightly positive in all conditions, suggesting that students tended to produce somewhat longer essays. Further, the notably large standard

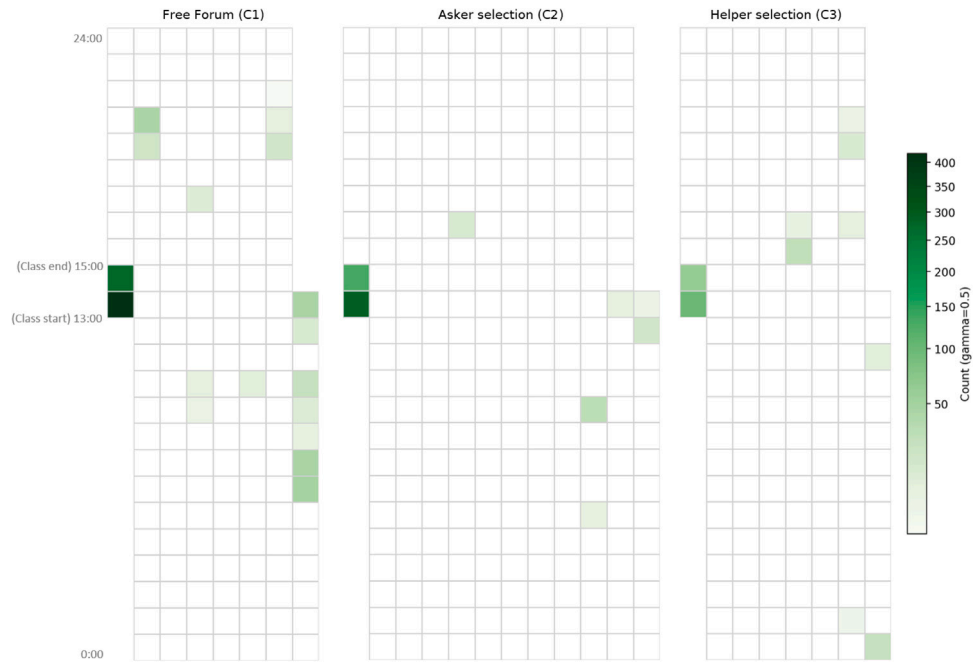


Fig. 8. Hourly engagement of BookRoll during the “peer review and revise” periods across three conditions.

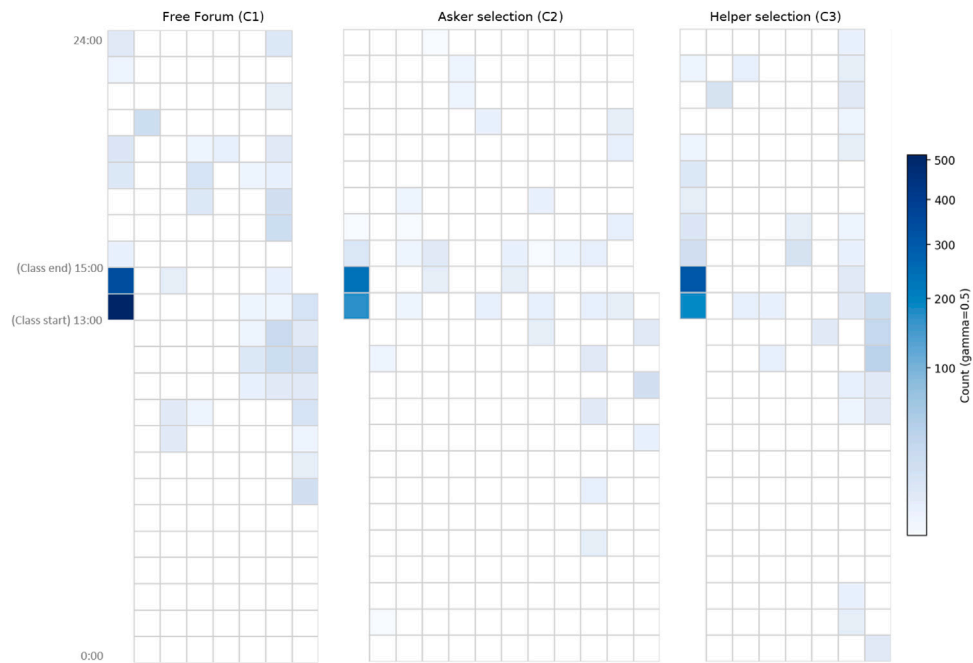


Fig. 9. Hourly engagement of peer learning during the “peer review and revise” periods across three conditions.

Table 3
Normalized learning gain (LG) across three conditions.

Condition	Mean	Median	SD	<i>N</i>	Kruskal–Wallis <i>H</i>	<i>p</i>	ϵ^2
C1 (Free forum)	0.46	0.48	0.34	23	4.34	.114	.035
C2 (Asker selection)	0.30	0.14	0.35	24			
C3 (Helper selection)	0.20	0.13	0.51	24			

Note. LG = (Revised report score – Initial report score)/(Full score – Initial report score) [61]; ϵ^2 indicates the effect size for the Kruskal–Wallis test.

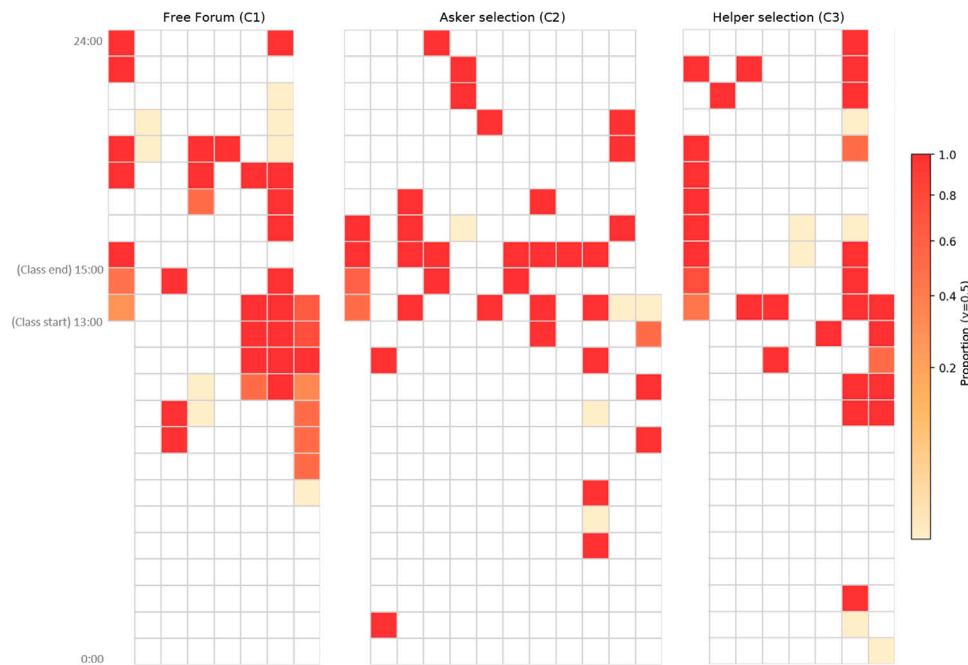


Fig. 10. Hourly proportions of the number of students who only engaged in peer learning, during the “peer review and revise” periods across three conditions. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

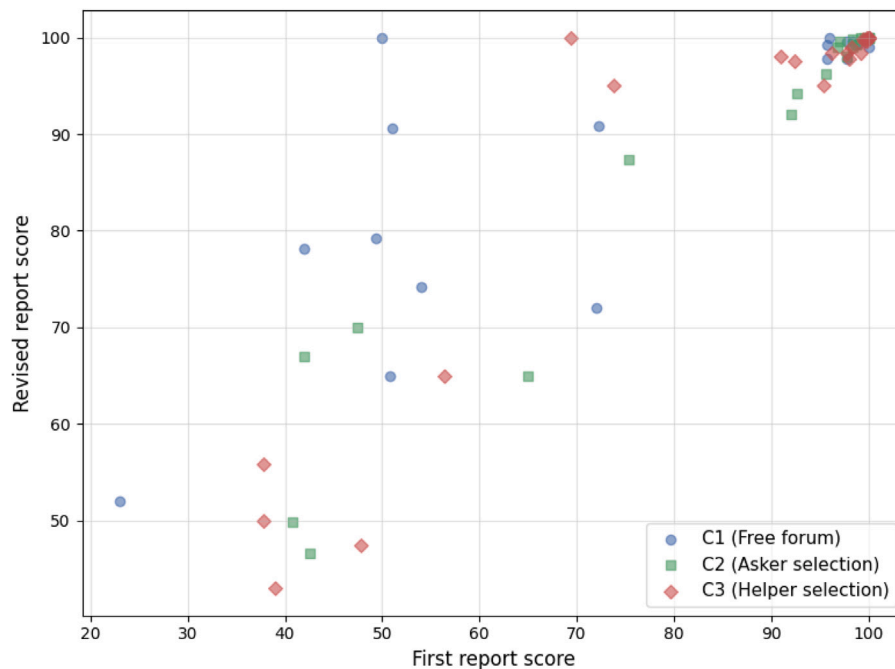


Fig. 11. Scatter plot of initial and revised report scores across three conditions.

deviation in C1 indicates greater variability in word-count improvement under the free forum-based condition, implying that students varied more widely in their degree of engagement or productivity when partner selection was unconstrained.

6. Discussion

6.1. Interpretations of key findings

Regarding the antecedent of peer learning (RQ1), namely how partners are determined, the results revealed that in the free forum

condition, students tended to choose partners based on existing social relationships. This finding aligns with prior studies showing that social intimacy strongly influences partner selection in collaborative settings, as they are less likely to form new ties and links to others [28]. In contrast, the data-driven system appeared to guide partner selection toward knowledge proficiency gaps, suggesting its potential to support more pedagogically meaningful pairing [46]. Within the system, knowledge gaps served as a social reference frame that guided learners' decisions, shaping their perception of suitable partners [49]. Together, these findings support the premise of this study that the three conditions represent distinct partner selection modes driven by

Table 4
Change in word count across conditions.

Condition	Mean Δ	Median	SD	N	Kruskal–Wallis H	p	ϵ^2
C1 (Free forum)	8.73	22.00	219.58	26	3.90	.142	.028
C2 (Asker selection)	18.16	9.00	88.09	25			
C3 (Helper selection)	55.54	32.00	116.02	24			

Note. Δ = (Final report word count – First draft word count); ϵ^2 indicates the effect size for the Kruskal–Wallis test.

different factors: social intimacy in the free forum and knowledge awareness in the peer learning platform.

The results for the processes and consequences of peer learning (RQ2 and RQ3) indicate that C2 and C3 increased efficiency by reducing the effort required to complete the peer learning tasks, while achieving learning outcomes that were not significantly different from those in the control condition. Students in C3 tended to show relatively larger increases in word count than those in the other two conditions, whereas changes in text length do not necessarily indicate deeper learning or improved writing quality. In terms of engagement, the results showed a decline that reflects the role of the data-driven digital system in conserving effort and improving the efficiency of teaching and learning. With respect to peer learning engagement, the free forum condition yielded the greatest number of operations, as students were able to browse all classmates' posts, which may have distracted them from focusing on their own tasks.

Further, even in spontaneous pairing condition (C1), social intimacy or friendship did not guarantee effective peer learning or greater task engagement, as trust and affective ties may operate in social or emotional domains rather than in academic collaboration performance [62]. According to the social interdependence theory by Johnson and Johnson [63], boundary interdependence based on proximity or similarity does not necessarily contribute to critical thinking or knowledge construction. Means interdependence that focuses on resource and task may play a more significant role. The results show that knowledge-based recommendation alleviated the effect of social intimacy in peer selection by focusing attention on knowledge asymmetry.

The process data analysis further disclosed the underlying reasons for this finding. Students in data-driven conditions tended to complete peer review and revision activities more directly within the peer learning platform. This pattern suggests that when proficiency gaps were large under a knowledge-driven condition, askers relied less on re-checking the original article during the revision phase. Because helpers were already familiar with the paper through their previous reading or reporting experience, they could provide adequate insights, allowing askers to comprehend and act on the feedback. As a result, the additional effort required for rereading the paper was reduced. This tendency was more evident in C3, where helpers held the autonomy to inspect knowledge gaps and determine which papers to review based on their preferences. Since the helpers occupy the role of knowledge disseminators with high proficiency, their metacognitive monitoring of their own strong performance may be more accurate and effective in the educational context than the metacognitive monitoring of lower-proficiency learners [64].

In practice, partner selection supported by recommendation and visualization reduced the effort of manual checking, helping to lighten the workload and maintain quality, which may mitigate burnout among university students [65]. However, it may also decrease motivation for extensive reading beyond the assigned tasks. At the same time, when focusing on the consequences, the results indicated that a knowledge-based recommendation alone did not enhance quality, and targeted instructional intervention in the peer review process remains necessary to improve it. Peer selection initiated by helpers embraced potential for higher efficiency. Overall, these findings highlight the importance of focusing on the learning process rather than solely on outcomes, supporting the metacognitive-driven framework by Janssen and Kirschner [56] to inspect collaborative learning in an overall perspective.

6.2. Implications

From an empirical perspective, this study illustrates how data-driven peer learning design can foster meaningful interaction and engagement in academic activities within a higher education context, grounded in observed behavioral patterns and learners' self-reported perceptions. Learners reported gaining new insights from diverse perspectives rather than pursuing a single standard answer, which is particularly valuable in higher education contexts that emphasize human creativity in the age of AI [66] and aligns with the theoretical perspectives of the ZPD and constructivism.

From a methodological perspective, another implication concerns the value of the experimental design with three conditions. By comparing free forum pairing with data-driven selection initiated by askers or helpers, the study provides a comparative framework that enables the examination of both social intimacy and partner-selection autonomy within the same course context. The study analyzes peer learning across social, behavioral, and learning outcome levels. This broader analytic scope helps capture peer learning dynamics in a more comprehensive manner and connects the observed patterns to the theoretical factors guiding peer learning.

From a practical perspective, although many recommendation algorithms have been developed based on knowledge gaps and prior learning logs with gradient levels [22,40], few have been embedded in authentic learning environments as daily tools for educational intervention with empirically grounded narratives. The present study integrates the OKLM learner model with peer learning tasks in an authentic course, bridging computational modeling and pedagogical design in a real educational context. It demonstrates the potential of such data-driven designs in higher education and highlights broader avenues for improvement by considering the influence of social intimacy and autonomy in partner selection.

Regarding broader educational applications, though this study implemented peer learning within a structured peer review activity, it also admits certain contextual limitations such as restricting each learner to one asker–helper pairing per round and excluding multiple Q&A iterations. Nevertheless, grounded in the GLOBE framework [26], this approach can be flexibly extrapolated to broader contexts, including causal questioning and answering activities and secondary school settings. The peer interactions are not limited to textual forums but can also occur through oral communication or within virtual spaces [67,68]. In this sense, transferability across age groups, subjects, and communication modes is realistic provided appropriate data sensors can capture the antecedents, processes, and consequences of peer interaction as analyzable data. Under this condition, the learning content and learner features can be meaningfully represented within the OKLM schema, thereby enabling the GLOBE-informed peer learning framework demonstrated in this study to be applied to broader scenarios.

6.3. Limitations

This study has several limitations. First, the sample size was relatively small, though it is representative of small-scale courses that have become increasingly common in universities in recent years. Student absences were also frequent, and only cases that completed the full round of peer learning activities were analyzed, which led to the exclusion of several incomplete cases.

Second, the longitudinal effects and continuous updating of the learner model were not sufficiently realized, which limited the extent to which each student could demonstrate diverse strengths across different papers and fully orchestrate a ZPD-based learning scenario. Though we adopted normalized learning gains measurement, the ceiling effect observed in the later rounds may still have influenced the results. As described in [Appendix A](#), this model aggregates behavioral evidence from markers, reports, and reviews into proficiency scores and combines different evidence sources using assigned weights. As these score assignments and weight settings were based on heuristic operational choices, they may not precisely capture learners' underlying knowledge states. The findings should therefore be interpreted with this measurement limitation in mind.

Third, this study focused primarily on report scores and word-count changes, which may not fully capture deeper forms of learning or conceptual understanding. Broader social and interpersonal effects, such as impacts on learners' well-being and sense of belonging, also deserve further investigation.

Fourth, the peer-learning structure was relatively constrained, as students participated in only one asker-helper pairing per round and did not have opportunities for multiple iterations of questioning and feedback. This design may limit the extent to which the findings reflect more authentic or sustained peer-learning processes.

Fifth, while the proposed system was designed to temper the influence of non-cognitive factors by recommending anonymous partners based on objective proficiency gaps, such factors cannot be fully eliminated. Learners' motivation, willingness to interact, or sense of agency may still affect how actively they engage with an assigned partner.

Finally, there were inherent limitations in the digital data sensors used. Some learning activities may have occurred in external systems beyond BookRoll and the peer learning platform, such as through generative AI tools for automatic summarization, as reported by some students. Since the use of such external systems was not systematically monitored or analyzed, it may have influenced both learning processes and outcomes, and the behavioral logs analyzed in this study may not fully capture the complete picture of learners' study patterns. Future studies should explicitly monitor and analyze the use of external AI systems through self-reports, activity logs, or experimental control conditions to better understand how such tools may influence participant performance and the validity of the results.

At the end of the day, the limitations above suggest that the conclusions should be interpreted as context-specific evidence of potential efficiency and partner-selection benefits, rather than as broadly generalizable claims about the effectiveness of data-driven peer learning systems.

7. Conclusion

Collectively, the present study demonstrates the applicability of a data-driven peer learning system in partner selection and reducing social bias. Empirically, students in the free forum condition tended to choose partners based on existing social ties, whereas the recommendation and visualization functions guided partner selection toward knowledge proficiency gaps. Based on the observed behavioral patterns, the integration of recommendation and visualization functions can be interpreted as supporting greater efficiency in the peer review process, while no statistically significant differences in learning outcomes were observed across conditions, hence indicating its potential to balance workload and learning quality in higher education contexts. Helper-initiated pairing further showed relatively higher efficiency when learners could autonomously identify knowledge gaps and select partners.

These findings are significant for peer learning research as they indicate that partner selection is not only a logistical arrangement but also a process shaped by social intimacy, proficiency gaps, and selection autonomy. Though knowledge-based algorithmic recommendations did

not inherently improve learning quality, their pedagogical value further depends on instructional guidance that stimulates reflection and metacognitive engagement, positioning peer learning as a constructive process rather than a task-oriented exchange. This study also extends LA research from primarily teacher-centered or learner-centered support toward peer-centered learning by showing how interpretable recommendations and visualizations can be embedded in authentic course activities. Future studies should implement continuously updated learner models and inspect long-term effects across iterative peer interactions. Additional research could also explore multimodal extensions, such as incorporating verbal and virtual interactions, to capture richer collaborative dynamics and extend applications to diverse educational settings, thereby fostering sustainable learning, cultivating soft skills and human creativity in the age of AI.

CRedit authorship contribution statement

Changhao Liang: Writing – review & editing, Writing – original draft, Validation, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Yu-Tung Chen:** Writing – review & editing, Software, Methodology, Data curation. **Yu Yan:** Writing – review & editing, Methodology, Data curation. **Hiroaki Ogata:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Informed consent

This research work includes human participants and their participation has been vetted and approved by the Experiment Ethics Committee of Kyoto University.

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Declaration of competing interest

The authors declare that they have no conflict of interest.

Appendix A. Details of the OKLM-based recommendation mechanism

This appendix provides a detailed description of the OKLM-based recommendation mechanism and explains how raw behavioral data were transformed into quantitative proficiency scores. The overall design follows the data-driven peer help system introduced by Jiang et al. [54] and the recursive learner-model updating logic described by Liang et al. [55]. In the present study, this logic was adapted to the academic reading course by defining knowledge nodes as papers and their topic-level relationships.

A.1. Knowledge and learner models

As illustrated in [Fig. 1](#), the OKLM schema consists of two types of nodes. Knowledge nodes (*K*) represent learning units embedded in resources, whereas learner nodes (*L*) represent individual students. Edges among knowledge nodes constitute the knowledge model and indicate semantic, topical, or resource-based relationships within the domain. Weighted links between learner nodes and knowledge nodes constitute the learner model, and each link weight represents the

proficiency value $P_{l,k}$, where l denotes a learner and k denotes a knowledge node. Thus, OKLM does not simply store raw activity traces; rather, it transforms these traces into interpretable learner-knowledge relationships that can be used for recommendation and visualization.

In the general OKLM updating logic, raw behavioral logs are first mapped to the corresponding knowledge nodes. Individual learning activities, such as reading and annotation in BookRoll, contribute to individual learning gains $LG^{(I)}$, whereas peer-learning activities, such as writing reports and reviewing peers' reports, contribute to peer-interaction gains $LG^{(P)}$. The proficiency value is therefore updated as:

$$P_{l,k} = w_I LG_{l,k}^{(I)} + w_P LG_{l,k}^{(P)},$$

where w_I and w_P are adjustable weights for individual and peer-learning evidence, respectively. This update allows proficiency to change recursively as students alternate between asking and helping roles.

A.2. Recommendation scoring

When a peer-learning request is created, the selected papers or topics define the target problem space Q . For each candidate learner l , the system retrieves the learner's connected knowledge nodes K_l and their proficiency values $P_{l,k}$. It then calculates the graph distance between each candidate knowledge node $k \in K_l$ and each target node $q \in Q$. The recommendation score is computed as:

$$S_l(Q) = \sum_{q \in Q} \sum_{k \in K_l} \frac{P_{l,k}}{D_{k,q}},$$

where $D_{k,q}$ denotes the distance-discounting term derived from the shortest path between k and q in the knowledge graph. A smaller distance indicates stronger conceptual relevance to the target problem space. Candidates with higher $S_l(Q)$ values are ranked higher because they combine stronger proficiency with closer relevance to the selected knowledge nodes.

A.3. Calculation of proficiency indicators in this study

In the present study, the general OKLM proficiency value $P_{l,k}$ was operationalized as a raw proficiency score ranging from 0 to 1. Each behavioral record was first mapped to the corresponding paper node and then aggregated from three sources of evidence:

- *Annotations from markers:* BookRoll marker logs were used as evidence of individual reading. For each paper, the annotation contribution was calculated as $\pm 1/60 \times$ the number of markers assigned to the relevant words: $+1/60$ for highlight markers and $-1/60$ for not-understood markers. The denominator was set with reference to the first-half reading logs, where approximately 20 markers were typically observed in a carefully read paper.
- *Papers reported:* A learner received $1/3$ point for each paper on which the learner posted a report.
- *Papers reviewed:* Review-related scores ranged from $1/30$ to $1/6$ according to peer ratings. Intermediate values were calculated as $1/30 \times$ the peer rating score (1–5). If a review was posted but not evaluated, a default score of 0.1 ($1/30 \times 3$) was assigned; if no review was posted, the score was 0.

The resulting raw proficiency score for learner l on paper node k can be expressed as:

$$P_{l,k} = A_{l,k} + R_{l,k} + V_{l,k},$$

where $A_{l,k}$ denotes reading annotation-based evidence, $R_{l,k}$ denotes report-writing evidence, and $V_{l,k}$ denotes review-related evidence. Topic-level proficiency was then derived by aggregating the paper-level

proficiency scores according to the paper-topic assignment in the two-layer knowledge model. The overall OKLM structure instantiated in this study is summarized in Fig. A.1.

Appendix B. The rubrics for report grading and prompts for ChatGPT-4o

This appendix details the comprehensive evaluation framework used to assess student reports, ensuring consistency between manual (teacher) and automated (GPT-4o) grading. It is divided into two sections: the detailed grading criteria (B.1) and the exact prompts used for the automated scoring system (B.2).

B.1. Grading criteria

The following criteria were established for the evaluation of all student reports.

B.1.1. General mandate

All submissions MUST be written in English. Submissions in any other language will receive a score of 0.

B.1.2. General evaluation criteria (Applies to both summary and discussion)

For any key point to be considered fully met, it was evaluated against the following three general criteria:

- Is the writing clear and understandable, even for someone who has not read the original paper?
- Did the writer provide enough relevant information from the paper to support their statements or opinions?
- Did the writer cover all the important points without going into unnecessary detail?

B.1.3. Scoring scheme

1. Summary (Maximum Score: 50 points) The summary was required to include the following key points. (Note: If a criterion was not applicable to a specific paper, it was considered as having been met.)

• Key Points Required:

- Did the writer identify the main research question(s) addressed in the paper?
- Did the writer highlight the core contribution or innovation of the study?
- Did the writer describe the experimental procedures, techniques, or theories used in the research?
- Did the writer explain or interpret the results presented in the paper, including how the data were analyzed?
- *Bonus Point:* If they explained how this research fills a gap which was not addressed in previous studies.

• Scoring Logic for each Key Point:

- If the student mentioned the point but met 0 of the general criteria $\rightarrow +5$
- If the student mentioned the point and met 1 of the general criteria $\rightarrow +10$
- If the student mentioned the point and met 2 of the general criteria $\rightarrow +12$
- If the student mentioned the point and met all 3 of the general criteria $\rightarrow +15$
- (Exception: Some key points, such as the main RQ, might only be able to meet 1 criterion, resulting in a max of +10).

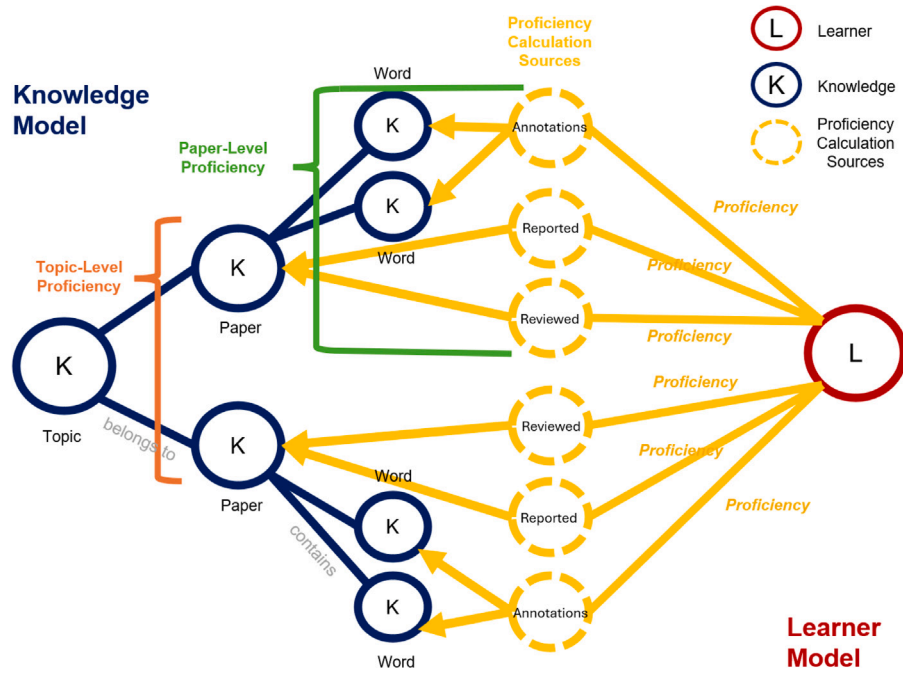


Fig. A.1. Overall OKLM structure instantiated in this study.

2. Discussion (Maximum Score: 50 points) For the discussion, originality, effectiveness, and readability were evaluated separately based on the following considerations.

• **Considerations for Evaluation:**

- If the writer mentioned the relation of the paper to the field of learning analytics, was the relationship clearly explained?
- If the writer commented on the paper's originality, effectiveness, readability, etc., did they provide clear reasons or evidence to support their views?
- *Bonus Point:* If they offered their own ideas or insights (e.g., suggestions for improvement and critical thinkings).

• **Scoring Logic for Discussion:**

- The student must write at least 3 discussion points to be eligible for the full score (questions are also considered discussion points).
- For each key point mentioned but not meeting criteria → +10
- For each key point mentioned that also met relevant criteria (see B.1.2) → +15
- For the bonus point → +5

1. **Role-Setting:** Instructing, “You are a professional university teaching assistant”, primes the model to adopt an expert, academic tone.
2. **Task Definition:** The task is unambiguously defined: “to grade a student’s assignment based on the provided Original Paper and detailed Grading Criteria”.
3. **Process Guidance:** “You must think step by step, calculate the score precisely”, is a form of process guidance (akin to Chain of Thought) to improve analytical accuracy.
4. **Output Constraint:** The final command, “return ONLY a JSON object”, is a critical constraint to ensure the output is strictly machine-readable.

This image shows the initial section of the user_prompt (Fig. B.3). The primary idea here is to establish the complete context for the AI. It is first shown the Homework Requirement so it understands the student’s task. It then defines three distinct material inputs using placeholders: {original_paper} (Material 1: The source text), {content_to_grade} (Material 2: The student submission), and {grading_criteria} (Material 3: The rules for scoring). This separation of materials is crucial for clarity.

This image displays a critical component of the prompt (Fig. B.4): a “few-shot” example (Example 1). The purpose of this section is to calibrate the AI’s judgment. By providing a concrete pair of student text and a detailed Evaluation Score (the “ground truth”), the model learns the *application* of the grading criteria, not just their abstract definition. This example demonstrates the expected granularity of scoring (e.g., RQs: 10, novelty, contribution: 15) and helps align the AI’s output with the teacher’s evaluation style.

This final image (Fig. B.5) details the most critical instructions. First, it outlines the task (Your Task and Instructions), instructing the AI to “think step by step”. Second, it defines the Output Result format with extreme precision. The core idea is to ensure a machine-readable, parsable output. By providing a commented jsonc template, we force the AI to return **only** a JSON object matching this exact schema. The comments inside the schema (e.g., // Score range: 0–10, // **CRUCIAL STEP**...) serve as in-line instructions, guiding the AI on score ranges and the logic for calculating capped totals.

B.2. GPT-4o prompting for automated scoring

To implement automated grading, we provided the GPT-4o model with a structured, multi-part prompt. Since this prompt is complex and includes few-shot examples and specific output format constraints, we provide a detailed breakdown through the following series of screen-shots (Figs. B.2 to B.5).

This image shows the system_prompt (Fig. B.2), which serves as the foundational instruction for the AI model. The core idea behind this prompt is to establish a clear “persona” and “objective” before any user-specific data is processed. This is achieved through four key instructions:

```

system_prompt = """
You are a professional university teaching assistant.
Your task is to grade a student's assignment based on the provided Original Paper and detailed Grading Criteria.
You must think step by step, calculate the score precisely,
and finally return ONLY a JSON object containing the final score.
"""

```

Fig. B.2. Prompt part 1: System prompt (Role and task definition).

```

user_prompt = f"""
Please grade the student's assignment based on the following three materials.

Homework Requirement:
Summary (write all of the following)
1. What is the research question of this paper?
   What kind of question does the paper raise and investigate?
2. What is the novelty or main contribution of this paper?
   Compared to other studies, what are the new aspects or particular contributions?
3. What experimental procedures, techniques, or theories were used?
   What kinds of methods, technologies, or theoretical frameworks were applied in the paper? Please organize and summarize them.
4. What results were obtained, and how were they analyzed?
   What findings were presented, and in what way were they analyzed?

Discussion (write as much as possible)
• Scope
  How is this study related to the field of Learning Analytics?
• Originality
  Do the research question or methods present novelty? Is there clear differentiation from existing studies?
• Effectiveness
  Are the methods and data sufficient to support the results? Were there any logical inconsistencies or unacknowledged limitations?
• Readability
  Which parts were difficult to understand? Was the paper's structure and flow easy to follow?
• Others
  Like in the "Archie paper," try asking further questions. Was there anything that especially made you want to ask the authors?

### Material 1: Original Paper ###
{original_paper}

### Material 2: Student Submission to be Graded ({content_name}) ###
---
{content_to_grade}
---

### Material 3: Grading Criteria ###
---
{grading_criteria}

```

Fig. B.3. Prompt part 2: User prompt (Context definition and input placeholders).

```

Example 1:
1. Better interfaces in the field of education are examined by comparing spreadsheets and GUIs at various evaluation points.
2. Until now, the best interface for editing texts and questions in education has not been seriously considered, but this paper explores
3. Thirty-eight educators were asked to edit algebra problems using a spreadsheet and GUI, respectively, to quantify the time taken to
4. GUIs are superior to spreadsheets in terms of speed and ease of editing when using single options (e.g. tips, diagrams, graphs), while
Originality: The paper is considered original in this respect, as it states that no serious consideration of the interface has ever been

Evaluation Score:
- RQs: 10
- novelty, contribution: 15
- method: 12
- result: 12
- bonus1: 0
- scope: 15
- ordinary: 15
- effectiveness: 0
- readability: 0
- others: 0
- bonus2: 0
- summary: 49
- discussion: 30
- all: 79

```

Fig. B.4. Prompt part 3: Few-shot calibration example.

```

### Your Task and Instructions ###
1. First, think step by step. Carefully read the Original Paper and the Grading Criteria.
Analyze the "Student Submission to be Graded ({content_name})" against each criterion.

2. **Output Result**:
- Your final output MUST be ONLY one JSON object that strictly follows the structure and schema shown in the template below.
- The comments in the template (marked with '//') are instructions that you must follow for assigning values.
- Do NOT include the comments or any other explanations in your final JSON output.
- All score values must be integers.

```jsonc
// --- JSON OUTPUT TEMPLATE ---
{
 "scores": {
 "summary": {
 "RQ": <integer_score>, // Score range: 0-10
 "novelty_contribution": <integer_score>, // Score range: 0-15
 "method": <integer_score>, // Score range: 0-15
 "result": <integer_score>, // Score range: 0-15
 "bonus1": <integer_score> // Score range: 0-5
 },
 "discussion": {
 "scope": <integer_score>, // Score range: 0-15
 "originality": <integer_score>, // Score range: 0-15
 "effectiveness": <integer_score>, // Score range: 0-15
 "readability": <integer_score>, // Score range: 0-15
 "other": <integer_score> // Score range: 0-15
 "bonus2": <integer_score> // Score range: 0-5
 },
 "total": {
 "summary_total": <integer_score>, // **CRUCIAL STEP**: First, sum the 5 scores in "summary".
 // If the sum > 50, this value MUST be 50. Otherwise, use the calculated sum.
 "discussion_total": <integer_score>, // **CRUCIAL STEP**: First, sum the 6 scores in "discussion".
 // If the sum > 50, this value MUST be 50. Otherwise, use the calculated sum.
 "grand_total": <integer_score> // **CRUCIAL STEP**: Calculate (summary_total + discussion_total + bonus).
 // This final value MUST NOT exceed 100.
 }
 }
}

```

Fig. B.5. Prompt part 4: Actionable instructions and strict output schema.

## Availability of data and material

Data supporting the results presented in this article are available from the corresponding author upon reasonable request.

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